

# Digital diplomacy and measurement of electronic public data qualities

## What lessons should be learned?

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### Abstract

**Purpose** – This paper aims to present a recent study on the definition and measurement of quality dimensions of public electronic records and archives (QADEPs: Qualités des archives et documents électroniques publics). It develops an original model and a complete method with tools to define and measure electronic public data qualities within public institutions. It highlights also the relationship between diplomacy principles and the measurement of trustworthiness of electronic data in particular. This paper presents a general overview of the main results of this study, with also illustrative examples to demonstrate the feasibility of measuring the qualities of electronic archives in the context of public institutions.

**Design/methodology/approach** – This research was conducted in two phases. The first one was the conceptual phase in which the quality dimensions were identified and defined with specific sets of indicators and variables. The second phase was the empirical phase which involved the testing of the model on real electronic documents belonging to several public institutions to validate its relevance and applicability. These tests were performed at the Archives of the State of Wallis and the Archives of the State of Geneva, thanks to different measurement tools designed especially for this stage of the research.

**Findings** – The QADEPs model analyzes the qualities of electronic records in public institutions through three dimensions: trustworthiness, exploitability and representativeness. These dimensions were divided into eight sub-dimensions comprising 17 indicators for a total of 46 variables. These dimensions and their variables tried to cover the main aspects of quality standards for electronic data and public documents. The study demonstrates that nearly 60 per cent of the measured variables could be automated.

**Research limitations/implications** – The QADEPs model was defined and tested in a Swiss context on a limited sample of electronic public data to validate, essentially, its feasibility. It would be useful to extend this approach and test it on a broader sample in different contexts abroad.

**Practical implications** – The decisionmaking of records retention in organizations and public institutions in particular is difficult to establish and justify because it is based generally on subjective and non-defendable practices. The QADEPs model offers specific metrics with their related measuring tools to evaluate and identify what is valuable and what is eliminable within the whole set of institutional electronic information. The model should reinforce the information governance of those institutions and help them control the risks related to information management.



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**Originality/value** – The current practice of archival appraisal does not yet invest in a meticulous examination of the nature of documents that should be preserved permanently. The lack of studies on the definition and measurement of the qualities of electronic and public electronic records prevents verification as to whether archival materials are significant. This paper fills in some of the gaps.

**Keywords** Appraisal, Information management, Archives management, Electronic records management, Accuracy, Digital storage

**Paper type** Research paper

## Introduction

In archival science, the interest for studying the concept of quality is not well-developed. This paper presents the main results of QADEPs project (Qualités des documents et des données électroniques publics) realized at the Geneva Business School with the collaboration of several public and private Swiss partners, who are themselves dealing with the definition and the measurement of the important qualities applied to electronic records and data in public institutions. It specifies the measurement of three dimensions:

- (1) trustworthiness;
- (2) exploitability; and
- (3) representativeness.

For the purpose of this paper, the author focuses in particular on the main indicators and variables related to the trustworthiness dimension. Classic and digital diplomatics literature represents one of the main resources in this field.

The goal of the project is, in a conceptual phase, to define the variables to appraise the quality of public electronic archives in the perspective of a sustainable conservation in an Open Archival Information System (OAIS) system type. In a second empirical phase, the model is tested on three public institutions (Swiss Federal Archives, Archives of the State of Geneva and Archives of the State of Wallis) to verify its applicability. This research promises to bring significant benefits: theoretical results as well as empirical results. Regarding the theoretical results, the conceptual framework will be presented as a general mapping, which will summarize the dimensions and indicators of the qualities of electronic records. This structure will represent the various aspects of the quality of public electronic archives. A sample of variables related to authenticity will be then presented to enable an accurate understanding of how one can define and quantify the qualities of a public electronic document to ensure its long-term preservation. Moreover, this paper will focus the trustworthiness dimension of electronic records studied as part of the QADEPs project. In short, it will highlight in particular the measurement of authenticity and will demonstrate a relevant use case developed on the basis of rich diplomatics knowledge.

### *Relationship between diplomatics and the study of records qualities*

Diplomatics is recognized as a medieval discipline that consists of studying exclusive attributes of records and manuscripts. It focuses not only on the analysis of their nature but also on their broad context as well as their creation and their transmission

conditions (Duranti, 1998). What is studied in diplomatics in particular is the relationship between the archives (content, container and context) and the fact/act with which those archives are dealing. Digital diplomatics has the same interest, except that its scope is slanted toward an electronic environment and, more specifically, toward digital documents or those which were converted into a digital object from a paper document.

The main goal of diplomatics and digital diplomatics is to examine the intrinsic and extrinsic characters of records to demonstrate their authenticity. In this study, authenticity, reliability and other internal and contextual records attributes will be studied as important characteristics that should be considered when establishing the quality levels of those records. Thus, from this perspective, the author will take advantage of this old discipline to conduct an emergent and recent research which is focused on defining and measuring qualities of digital records.

### **QADEPs project: main lines**

This section presents the main lines of the study. The context and background will be presented first to allow a full understanding of the current issues linked to the assessment of the quality of electronic documents and, therefore, the need of a practical implementation of this project. Then, the three objectives of this research will be specified to give an insight into the guiding principles that were followed throughout the project.

#### *Context and background*

Whether in public or private companies, any good activity process contains one or more stages dedicated to the control of its quality, service and the product that results from this process. This phase is also considered as capital, as shown by two series of standards: ISO 9000 and ISO 14000. In September 2011, the International Organization for Standardization published a third series, ISO 30000, on records management activities. This demonstrates the importance of integrating quality principles in the implementation of information governance processes.

Documents produced or received by each employee as part of his/her duties at each stage of the activity processes should be well-documented. Therefore, it is necessary to first identify and define clearly and systematically the whole life cycle of each activity and its related processes, and second, to define the quality criteria that should be applied for selecting relevant information derived from all these institutional processes and activities.

The application of quality criteria will enable an adequate reduction of the mass of documents to scan, as only a small volume of them contains the key information that needs to be saved and stored. Digitizing services will thus greatly be facilitated, and the institutions will have more savings opportunities. This offers a pragmatic and methodological way to ensure the setting and control of the quality of digital information. The organization will, therefore, significantly increase the excellence of its information service. These arguments show the relevance of this project in the current context. Providing a conceptual framework and operational measuring instruments corresponds to the process of the information quality assessment by which an institution would ensure the rationality of its information, record and data management practices.

### Objectives

This project has three objectives:

- (1) Identify and define key dimensions of the quality of electronic and public documents and their related indicators.
- (2) Identify the variables and develop the appropriate and operational tools for assessing the quality of public and electronic documents.
- (3) Test the applicability of the measurement of each variable on real cases to verify its relevance, feasibility and its automation.

The realization of these objectives requires an exploratory approach which will be exposed in the next section.

### Methodology

This section is structured into two parts: the first one describes the conceptual phase of the study, which refers essentially to the definition of the conceptual framework. The findings of this particular phase have played a crucial role, as they influenced the overall structure of the study. The second section highlights the main stages of the development of the empirical phase in which the author developed the measuring process.

#### *Conceptual phase: definition of a conceptual framework*

The conceptual framework describes the dimensions and key indicators of electronic records quality in the context of public institutions. Through an extensive literature review, the goal was to provide a consistent model based on reliable resources (including international standards), to give a clear representation of what is encompassed by the notion of quality for electronic and public records. Developing concepts definitely helps reaching an adequate understanding of various situations, but it does not solve or control them. For this, more operationalization needs to be developed.

To manage the flow of electronic documents and to preserve their legal value, various projects (InterPARES, KEEP Project, etc.) and a number of standards (ICA-Req, ISO 14641, ISO 14721, Moreq, 2010) have been created and carried out since the 90s. However, none of these projects or texts has suggested a suitable solution to ensure the quality of the documents (Makhlouf Shabou, 2011).

The present study takes advantage of the results of the author's recent research (Makhlouf Shabou, 2011) and adapts them to the study of the qualities of electronic records. To do so, an in-depth literature review was conducted focusing on the international standards related to information processing as in the ISO 15489, 23081, 26324, 30300, 30301 or the Afnor NF Z 42-013 standard. Those texts are valuable resources because they evoked some records qualities however they did not propose operational definitions for measuring these qualities. On this basis, it was then possible to define the variables to confirm whether the given recommendations in those standards have been applied.

Among the other essential references, useful to the development of the author's analyzing tool, there were the practices of a number of institutions as the [National Archives \(2015a, 2015b\)](#) of various countries (especially Switzerland, Canada, Germany and Australia). The practices of the latest institutions provide a point of comparison with the standards by describing how document processing is conducted under a perspective of long-term archiving. Indeed, most of the standards considered here

assume rather a records management perspective (current and intermediary records), and once the life cycle period of the documents ends, the priorities in terms of archiving also evolve. For example, the use of encryption guarantees security during the life of the document; nevertheless, it becomes a significant risk when it comes to long-term archiving. Most institutions, like [Library and Archives Canada \(2006\)](#), refuse to take responsibility for such objects[1].

The author explored also the scientific literature and the results of certain research groups like InterPARES. The issues addressed were often very specific, and therefore, it required an important effort of adaptation to draw operational variables. The bibliography at the end of this article will show the various references used ([Figure 1](#)).

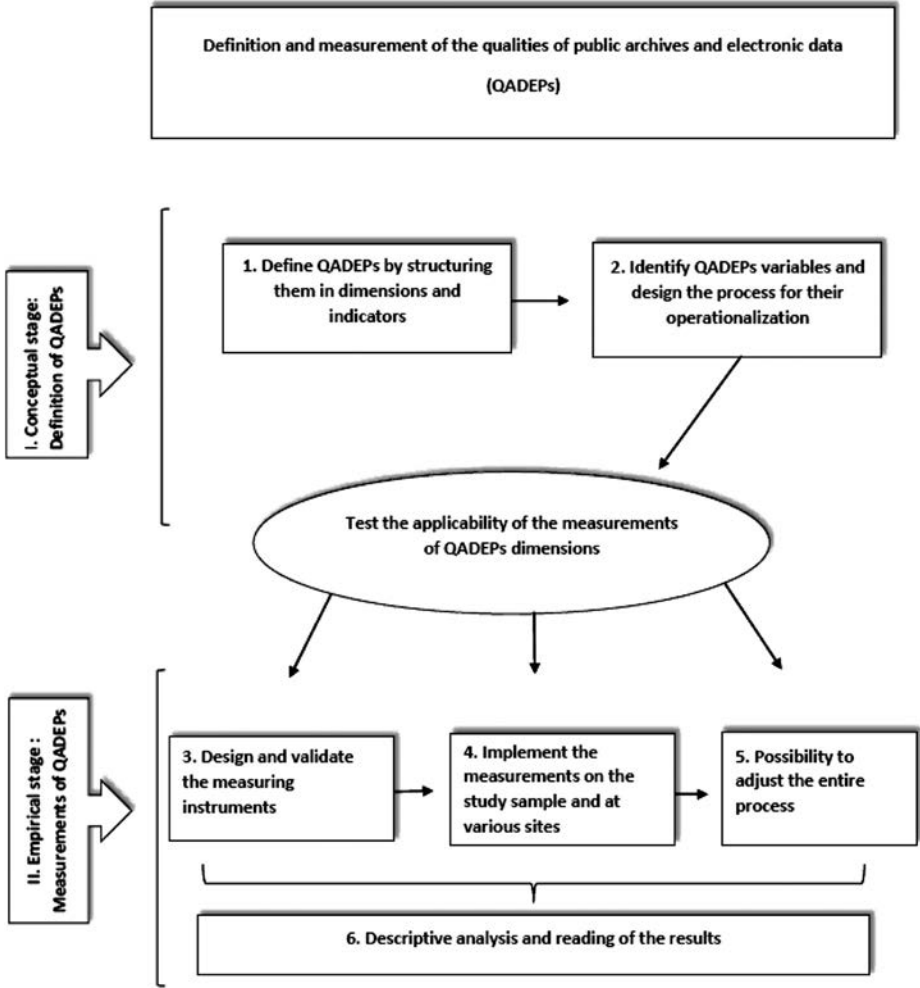


Figure 1.  
Methodology

By combining those various references, it allowed us to develop a reliable tool based on stable and recognized foundations (international standards), but also close to the institution practices and in compliance with existing research.

*Empirical phase: development of the measurement process*

Fortin (2006, p. 274) defines the operationalization, mainly, by a four-step process:

- (1) the development of the theoretical definition of the concept;
- (2) determining the dimensions of the concept;
- (3) the choice of empirical indicators; and
- (4) selection or development of the measuring instrument.

Drawing on this approach and based on the definition of intrinsic and contextual dimensions of QADEPs in terms of indicators, the author tested the operationalization of the model established on the measurement of a specific set of variables on a reduced sample of public and digital records.

The testing of the applicability of the QADEPs measurements was carried out in three steps:

- (1) designing of the QADEPs measuring instruments;
- (2) validation of instruments with different partners; and
- (3) implementation of the QADEPs measures on a sample.

For testing the identified metrics of the QADEPs project, the author selected three institutions:

- (1) the Swiss Federal Archives;
- (2) the Archives of the State of Wallis; and
- (3) the Archives of the State of Geneva.

These tests were primarily intended to check:

- *The relevance*: Do these metrics reflect a high level of quality for records and data?
- *The applicability of the variables*: Are they ambiguous? Is it easy to make a choice amongst the different quality levels?
- *The automation possibilities*: Would it be possible for a computer system to perform an automated quality measurement?

Once the conceptual framework was defined, the next goal was to develop variables to assess on measuring scales, the dimensions and indicators defined earlier. Four sub-objectives have guided the development and the verification of these measurements:

- (1) to enable automated calculation, which would offer an undeniable advantage for quality assessment of substantial quantities of data;
- (2) to limit as much as possible subjective assessments;
- (3) to document the application of each measurement to facilitate their reusability and their adaptability to other contexts; and

- (4) to confirm the applicability and relevance of the variables through tests in several institutions.

Overview of findings

This section explains the findings resulting from the two main phases of the project, the conceptual phase and the empirical phase. As an example, the author has chosen to develop the measurement of authenticity particularly to illustrate how the author worked on the automation question and to provide an example of the application tool.

Conceptual framework

The conceptual framework seeks to define the quality of electronic records according to several dimensions, sub-dimensions and indicators. It (Figure 2) has three dimensions and eight sub-dimensions which contain 15 indicators.

Dimensions and sub-dimensions

By dimension, the author means a particular facet of quality applied to electronic archives. For this study, the author selected three of them:

- (1) trustworthiness;
- (2) exploitability; and
- (3) representativeness.

Globally, the main dimensions proposed in this conceptual framework could be as defined in the following sections.

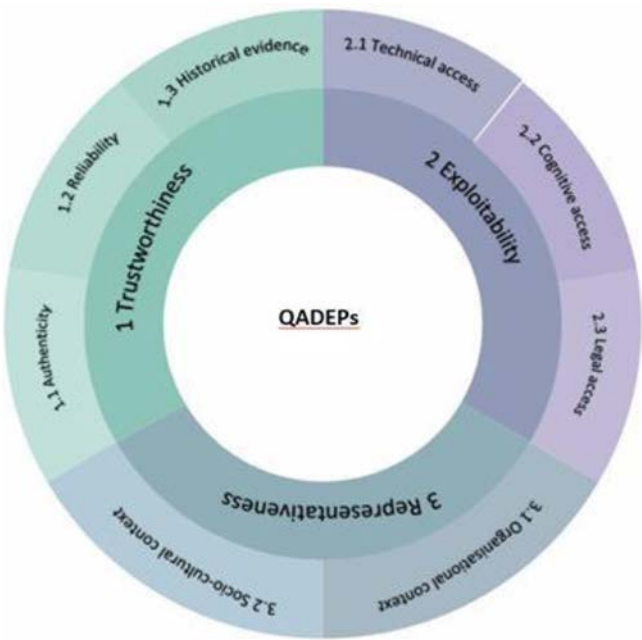


Figure 2.  
Overview of the  
conceptual  
framework



*Trustworthiness.* This dimension refers to the ability of a document to gain the trust of the user as the preferred supporting facts source. This quality depends on the authenticity and the reliability and the durability of these qualities over time (Makhlouf Shabou, 2011, p. 115; InterPARES 2, 2013).

*Exploitability.* This dimension refers to the ease of use of a document, thanks to its location, retrievability, diffusion and interpretability. In other words, the exploitability depends on three types of document accessibilities:

- (1) technical accessibility, including physical material needed for reading;
- (2) legal accessibility, including regulatory and administrative environments required for the diffusion of document; and
- (3) cognitive accessibility that guarantees an adequate comprehension and interpretation of document contents (ISO 15489, 2001; Makhlouf Shabou, 2011, pp. 120-122).

*Representativeness.* This dimension refers to the capacity of the documents to provide a significant testimony of the institutional context in which they were created. This quality depends on two essential elements:

- (1) the completeness of testimony; and
- (2) the representativeness of the socio-cultural context in which these documents were created (Makhlouf Shabou, 2011, p. 123).

### **Empirical phase: an overview of the main results with an illustrative case**

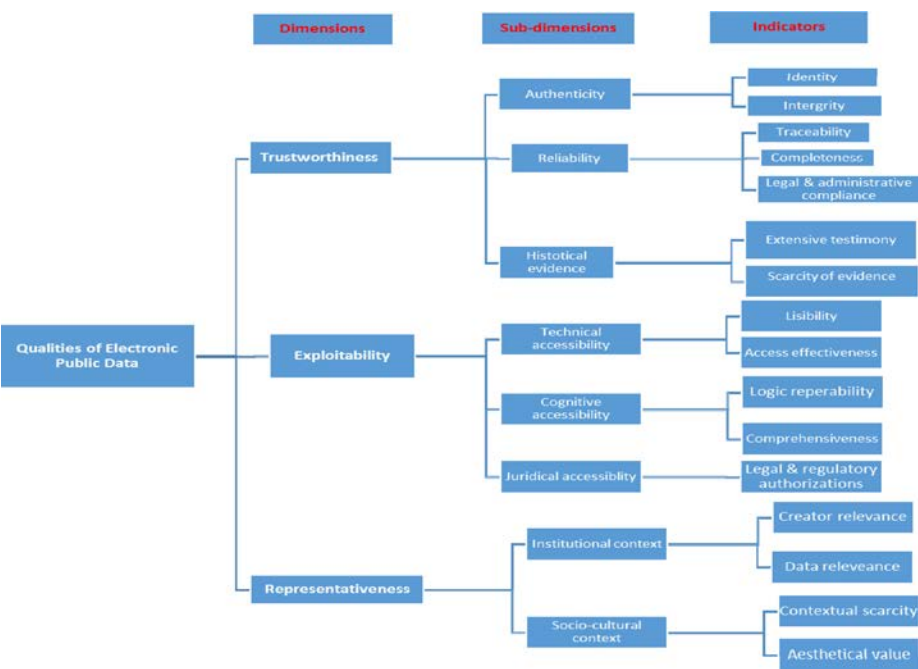
The development of the conceptual framework allowed an accurate definition of each concept deriving from the main quality dimensions. This was essential to identify adequately the variables and the metrics that will enable the measurement of these three dimensions. The realization of this last empirical phase allowed to define eight sub-dimensions detailed by 17 indicators, which are organized as shown in Figure 3.

The different qualities studied and tested in the context of electronic public data are structured as follows (Figure 3):

- (1) *Trustworthiness (dimension):*
  - Authenticity (sub-dimensions):
    - Identity (indicator).
    - Integrity.
  - Reliability:
    - Traceability.
    - Completeness.
    - Legal and administrative compliance.
  - Historical evidence:
    - Extensive testimony.
    - Scarcity of evidence.
- (2) *Exploitability:*
  - Technical accessibility:
    - Lisibility.
    - Access effectiveness.



**Figure 3.**  
Conceptual  
framework: structure  
of the dimension  
qualities



- Cognitive accessibility:
  - Logic reparability.
  - Comprehensiveness.
- Juridical accessibility:
  - Legal and regulatory authorizations.
- (3) *Representativeness*:
  - Institutional context:
    - Creator relevance.
    - Data relevance.
  - Socio-cultural context:
    - Contextual scarcity.
    - Aesthetical value.

The author will not detail the measurement of all dimensions presented here with their different hierarchical levels in this paper, but will focus more on the authenticity concept, which may be the most discussed concept in the literature, both in digital diplomatic and also in different international standards (see section *Representativeness*).

On the basis of the last conceptual level, which is the indicator level, variables were developed and tested to verify:

- their relevance;
- the feasibility of their empirical measurements; and
- the possibility of their automation.

### *Assessment method: quality levels and data scoring*

The 46 variables were identified for three quality dimensions and their related sub-dimensions and indicators. At least, for each indicator, the author provided a minimum of one variable or more to ensure its measurement. The measurement of each variable is based on a ranking system that provides specific scores reflecting the quality level of each evaluated data. The attribution of those scores was based on the verification of a set of specific conditions.

For example, to evaluate the integrity indicator of electronic data, the author looked at two variables:

- (1) *Access control*, which refers to the degrees of control, security and traceability of data access. Specifically, four quality levels were proposed:
  - *Level 1*: Lowest one, no access control during the life cycle.
  - *Level 2*: Existing access control during the life cycle by a user authentication.
  - *Level 3*: Existing access control during the life cycle by a reporting of events.
  - *Level 4*: Highest one, existing access control during the life cycle by an algorithmic system (checksums).
- (2) *Fixity of electronic data*, which is based on the stability of its carrier and its format. For measuring this stability, three assessment levels were proposed:
  - *Level 1*: Lowest one, applied when the fixity is not controlled.
  - *Level 2*: Medium one, applied when the fixity is controlled by, at least one checksum.
  - *Level 3*: Highest one, applied when the fixity is controlled, periodically, by planned checksums.

The number of quality levels for each metric is different; it varies between two and six levels.

It depends on the complexity of each variable. The rank corresponding to quality levels obtained after each completed measurement was considered as the official score and final result of the measurement. This scoring method starts with Level 1, which is the lowest one. This method does not provide a Level 0, as it seems difficult to assume, categorically, that the quality of information is inexistent.

### *Application units*

There were three main application units during the measuring process:

- (1) items;
- (2) series; and
- (3) fonds.

These units have been used according to the following definitions:

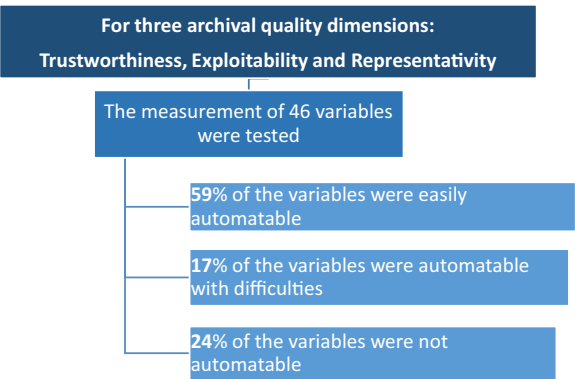
- *Items*: “A digital objects that represent a collection of data with defined boundaries that is treated as a single entity” (Pearce-Moses, 2014)[2].
- *Series*: “Group of similar records that are arranged according to a filing system and that are related as the result of being created, received, or used in the same activity; a file group; a record series” (Pearce-Moses, 2014)[3].
- *Fonds*: “The entire body of records of an organization, family, or individual that have been created and accumulated as the result of an organic process reflecting the functions of the creator” (Pearce-Moses, 2014)[4].

The series and fonds were considered, especially, in the measurement of the totality of the seven variables deriving from the *Representativeness* dimension, for which a large set of data was needed. These two units were also applied on the measurement of the five variables related to the sub-dimension of *Historical evidence* into the *Trustworthiness* dimension (Figure 3). For the rest of the variables, the author applied the items as measurement unit.

Each variable was measured by a specific metric. As shown in Figure 4, 59 per cent of the metrics were easily measured and could completely be automated. And also, 17 per cent of the metrics were computable, but would require the creation of specific metadata, which would be expensive. However, 24 per cent of the metrics were not computable due to the subjectivity of some criteria on which the assessment is based, as these are founded on manual and human evaluation. This reduces the systematic reproducibility of those metrics.

*A focus on authenticity and its measurement*

As mentioned above, authenticity represents an important sub-dimension in the *Trustworthiness* dimension. Authentic records are the ones that can prove they are what they are supposed to be, that they are really produced or received by the person who claims to have produced or received them (ISO 15489, 2001). At least, record authenticity is based on multiple types of matching metadata: the matching between its declared nature and its real nature; the matching of the declared producer/receiver and the real producer/receiver; and, finally, the matching between the declared creation/reception



**Figure 4.**  
Measurement of  
trustworthiness,  
exploitability and  
representativeness

date and the real creation/reception date. The efficient assessment of those previous conditions seems to be difficult; nevertheless, it ought to be possible if we try to examine some observable aspects related to records identity and integrity.

The identity is “The whole of the characteristics of a document or a record that uniquely identify it and distinguish it from any other document or record” (InterPARES 2, 2013). The measurement of records identity was based, according to the author’s study, on the examination of ten variables:

- (1) existing identifier;
- (2) existing title;
- (3) name format of records file;
- (4) history/journal of previous identifiers for a record;
- (5) history/journal of previous records title;
- (6) history/journal of previous file names;
- (7) knowledge of creator;
- (8) information about the file format;
- (9) information about timestamp; and
- (10) information about file size.

The integrity is:

[...]property of safeguarding the accuracy and completeness of assets. Integrity demonstrates that the record is complete and has been unaltered. It is necessary that a record be protected against unauthorized alteration. Records management policies and procedures should specify what additions or annotations may be made to the record after it is created, under what circumstances additions or annotations may be authorized, and who is authorized to make them. Any authorized annotation, addition or deletion to a record should be explicitly indicated and traceable (ISO 15489, 2001; ISO 13335-1:2004)[5].

The measurement of records integrity, was based, according to the author’s study, on the examination of three variables:

- (1) the rigor of the files and records access control;
- (2) fixity of the byte string; and
- (3) the recordedness of records in a specific system.

The measurement of authenticity, as one of the eight sub-dimensions developed in the author’s study, represents 13 of the 46 variables, which cover almost 30 per cent of the total number of variables. This demonstrates the importance of the place of authenticity in the electronic documents qualities assessment. While authenticity is not an exclusive interest to electronic documents but rather a quality that has been extensively studied in the context of diplomatics, it should definitely be recognized here as the real root of this research field. As shown in [Appendix 1](#), many sources propose a definition to this concept. They globally propose convergent definitions. However, they do not propose accurate methods to evaluate this quality. They also do not specify the variables that could be measured through this evaluation. Nevertheless, diplomatics sources, especially the InterPares research, briefly discussed the concept of authenticity assessment ([Table AI in Appendix 1](#)) without a deep description of its methods.

*Authenticity measurement: possibility of automation*

As mentioned earlier, once the relevance and the feasibility of the measure of authenticity was tested and validated, the author’s interest was directed to verifying the automation of those measures (Figure 5).

The measurement of variables related to the sub-dimension of authenticity was the most feasible. The testing of their automation was the easiest one because the metrics are based on the intrinsic qualities, which generally are given in the basic metadata made available by operating systems used for the basic processing of information.

*Authenticity measurement: the application guide*

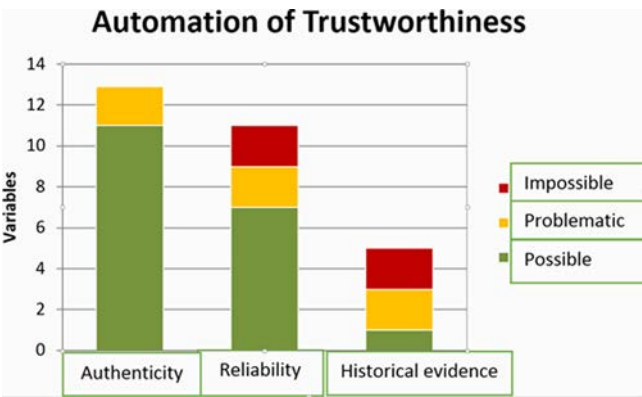
The aim of the application guide is:

- to provide a definition for the concepts that were used throughout the study;
- to offer a complete description for each dimension, sub-dimension, indicator and variable used and, therefore, to support the reproducibility of each measurement; and
- to justify the choice of different quality levels and to present the references on which the author’s study was based.

As shown in the sample presented in Figure 6, the overall content of the application guide follows the same structure as the general framework observed throughout the study. It begins with a definition of the dimension (in this study being trustworthiness, exploitability or representativeness), it continues with a definition of each sub-dimension and indicator selected, then one can find a description of the variables with their detailed application notes containing the explanation for each quality level chosen, and to conclude, the guide mentions the references on which the application notes were based. The application notes offer further information in the form of remarks or references mainly to specify certain application processes or to define a concept more accurately.

**Conclusion**

As mentioned earlier, the author’s research intended to define different dimensions that specify the qualities of data and electronic records in the context of public institutions



**Figure 5.**  
Automation of  
trustworthiness

**B. Detailed description of variables**

**1. Trustworthiness**

*Definition:* capacity of a document to gain his user confidence as being the privileged source in support of facts. This quality rests upon authenticity and reliability. (MAKHLOUF SHABOU, 2010, p. 115 ; INTERPARES 2, 2013)

**1.1. Sub-dimension : Authenticity**

*Definition:* « an authentic document is a document whose can prove that it is what it pretends to be, that it was produced or received by the person who pretends to have produced or received, and it was produced or received at the moment it pretends to have been. » (ISO 15489-1:2001, p. 7)

**1.1.1. Indicator : Identity**

*Definition:* « identity :The whole of the characteristics of a document or a record that uniquely identify it and distinguish it from any other document or record. With integrity, a component of authenticity. » (INTERPARES 2, 2013)

| v1. Existence of an identifier |                                                                         |
|--------------------------------|-------------------------------------------------------------------------|
| 1.                             | no existing identifier                                                  |
| 2.                             | existing identifier but without standards                               |
| 3.                             | existing and standardized identifier inside the institution             |
| 4.                             | existing and standardized identifier inside and outside the institution |

**Application note :** this variable examines the existence, the uniqueness, the standardization and the durability of the identifiers.

Level 1 : is applied when the object doesn't possess an indicated identifier in its current form.

Level 2 : is applied when an identifier exists, but there is no explicit rules about his establishment. This level takes, in most cases, the URI as their durability is not always guaranteed.

Level 3 : is applied when the identifiers are defined by the rules of the institution. The unicity is guaranteed, but only inside the institution. On the other hand we could find the same identifier attributed to another object in a different institution. It should be noted that the identifiers appealed to the ISIL codes attributed by a national center generally belong to this level because the ISIL code registers only the institution. The remainder of the identifier is managed and recorded internally.

Level 4 : is applied when the identifiers are defined according to norms or standards and are recorded in external registers of the institution. These identifiers must be definitively attributed and their unicity guaranteed. Their durability must be guaranteed, even if there are modifications or moving. Example: *Digital object identifier (DOI)*<sup>1</sup>, *Persistent URLs (PURL)*<sup>2</sup>, etc.

**Observations :** the same object can have several identifiers : for example, one for the internal functioning of the institution and another external DOI type. The assessment of the variable will only consider the identifier which has the highest value.

**Sources :**  
ORGANISATION INTERNATIONALE DE NORMALISATION. ISO 15511 : Information et documentation - Identifiant international normalisé pour les bibliothèques et les organismes apparentés (ISIL). Genève. 2011.

<sup>1</sup> DOI are made of a prefix and a suffix like the following : 10.xxxxxx/xxxxxx. The 10 is own to the DOI. The prefix indicate the agency which records the DOI, as for the suffix, it identifies the object it-self inside a register. The suffix can take numerous forms like an issn or any chain of characters.

<sup>2</sup>cf. PURL'S CONTRIBUTORS, 2011 : « PURLs are Web addresses or Uniform Resource Locators (URLs) that act as permanent identifiers in the face of a dynamic and changing Web infrastructure. Instead of resolving directly to Web resources (documents, data, services, people, etc.) PURLs provide a level of indirection that allows the underlying Web addresses of resources to change over time without negatively affecting systems that depend on them. This capability provides continuity of references to network resources that may migrate from machine to machine for business, social or technical reasons. »

**Figure 6.**  
The representation of  
the application guide:  
sample[6]

and their related sub-dimensions, indicators and variables. On the basis of those quality dimensions, a specific and rigorous measurement method was designed and tested.

Three main lessons should be learned from this study. First, the measurement of electronic records and data was possible, and its process was also verifiable and largely computable. Second, the method proposed in this study is reproducible. It is clearly presented. This study provides significant results which refer to a specific level of quality from the point of view of a particular aspect and cannot be understood as a general and absolute evaluation. For example, we can evaluate the exploitability of a given set of data as high-quality only from the point of view of specific sub-dimensions. Third, the realization of this research demonstrates the multidimensional nature of the definition and measurement of records and data qualities. Many elements should be studied and considered altogether to enable adequate definition and an accurate



measure of specific records or data quality. That is why the author noticed also the relative nature of obtained results specifically at the sub-dimensions and indicators levels which were based on the compilation and the interdependence between many variables.

In fact, the author's research, which defined and measured the quality of data and electronic records and archives, promises four significant benefits. In theory, it develops a conceptual framework that defines an accurate architecture of main qualities applied on electronic and public records and data. At the methodological level, it offers a method for measuring the historical qualities already tested in a real context of public institutions. At the professional level, first, it allows the assessment of the relevance of archival material that has been chosen for the appraisal; furthermore, it provides professionals a measurement grid for measuring archival qualities already tested and also the guide that facilitates its application.

At the academic level, this research has provided opportunities for collaboration with another department of the author's school. A group of five students from the Business Information Systems Department of the Geneva Business School are developing software that should enable the automation of the metrics related to authenticity and reliability of the sub-dimensions. The results are expected for the end of 2014. At the international level, the QADEPs results are shared with the InterPARES Trust research, and further collaborations are planned to develop this research area.

This study is, to the author's knowledge, the first reflection conducted in the domain of archival quality assessment. At this exploratory step, the author could not pretend to offer a way to measure exhaustively the quality of institutional archives. To do that, the research should be reproduced in different contexts to be confirmed and reinforced. The completion and the development of those measurements are necessary for the validity of this research's conceptual framework and the relevance of its results.

## Notes

1. [www.collectionscanada.gc.ca/gouvernement/produits-services/007002-3015-f.html](http://www.collectionscanada.gc.ca/gouvernement/produits-services/007002-3015-f.html)
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4. Definition available at: [www2.archivists.org/glossary/terms/f/fonds](http://www2.archivists.org/glossary/terms/f/fonds)
5. Available at: [www.ocio.gov.nl.ca/ocio/im/glossary.html#Integrity](http://www.ocio.gov.nl.ca/ocio/im/glossary.html#Integrity)
6. This is just one example of the contents presented in the guide. The English version of this guide with its exhaustive section will be available soon. Contact: [basma.shabou@gmail.com](mailto:basma.shabou@gmail.com) or [basma.makhlouf-shabou@hesge.ch](mailto:basma.makhlouf-shabou@hesge.ch)

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| Sources                                                       | Authenticity: definition and measurement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QADEPs                                                        | Indicator: identity<br>Existing identifiers for default format<br>Existing title for default format<br>Existing naming format for default format<br>Audit trail of existing identifiers for changes and/or changes in compliance with default format<br>Audit trail of existing titles for changes and/or changes in compliance with default format<br>Audit trail of existing naming formats for changes and/or changes in compliance with default format<br>Existence of metadata informing on the producer<br>Existence of metadata indicating the format<br>Existence of timestamp<br>Compare size of existing elements with size of previous elements<br>Indicator: integrity<br>Level of access control<br>Checksums control                                                                                                                                                                    |
| <i>Theoretical basis</i><br>Diplomatics:<br>InterPARES (2015) | Authenticity<br>n., The trustworthiness of a record as a record; i.e., the quality of a record that is what it purports to be and that is free from tampering or corruption. [Archives]<br><i>Assessments of authenticity</i><br>n., The determination of whether a document has all the formal elements that it was supposed to present when first made or received and set aside. [Archives]<br>Identity<br>Definition: the whole of the characteristics of a document or a record that uniquely identify it and distinguish it from any other document or record. With integrity, a component of authenticity. ≈<br>Source: <a href="#">INTERPARES, PROJECT, 2013</a> . Terminology database. InterPares Project [online]. 2013. [Consulted the 14th January 2015]. Available at: <a href="http://www.interpares.org/ip3/ip3_terminology_db.cfm">www.interpares.org/ip3/ip3_terminology_db.cfm</a> |
| ISO 14641                                                     | Integrity<br>Definition: attribute of a document whose content is completed and unaltered<br>Source: INTERNATIONAL ORGANISATION FOR STANDARIZATION, 2012. <i>Electronic archiving–Part 1: Specifications concerning the design and the operation of an information system for electronic information preservation</i> . Geneva: ISO, 24th January 2012. ISO 14641-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

(continued)

**Table AI.**  
Authenticity:  
theoretical basis

| Sources              | Authenticity: definition and measurement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OAIS: ISO 14721      | Fixity information: This information provides the data integrity checks or validation/verification keys used to ensure that the particular content information object has not been altered in an undocumented manner. Fixity information includes special encoding and error detection schemes that are specific to instances of content objects. Fixity information does not include the integrity preserving mechanisms provided by the OAIS underlying services, error protection supplied by the media and device drivers used by archival storage. The fixity information may specify minimum quality of service requirements for these mechanisms. Source: INTERNATIONAL ORGANISATION FOR STANDARIZATION, 2012. <i>Space data and information transfer systems –OAIS–Reference model</i> . Geneva: ISO, 21st August 2012. ISO 14721                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ISO 15489 (2001)     | Authenticité<br>Definition: Quality of a document can be proven to be what it claims to be, it was <i>effectivement produit ou reçu par la personne qui prétend l'avoir produit ou reçu, et qu'il a été</i> effectively produced or received by the person who claims to have produced or received, and was <i>produit ou reçu au moment o&amp;ugrave; il prétend l'avoir été</i> . produced or received by the time he claims to have been.<br>("un document authentique est un document dont on peut prouver qu'il est bien ce qu'il prétend &ecirc;tre, qu'il a été effectivement produit ou reçu par la personne qui prétend l'avoir produit ou reçu, et qu'il a été produit ou reçu au moment o&ugrave; il prétend l'avoir été")<br>Integrity<br>Definition: "Integrity is the property of safeguarding the accuracy and completeness of assets. Integrity demonstrates that the record is complete and has been unaltered. It is necessary that a record be protected against unauthorized alteration. Records management policies and procedures should specify what additions or annotations may be made to the record after it is created, under what circumstances additions or annotations may be authorized, and who is authorized to make them. Any authorized annotation, addition or deletion to a record should be explicitly indicated and traceable"<br>Source: <a href="#">International Organisation for Standardization (ISO) (2001)</a> |
| ICA-Req              | Authenticity<br>Definition: the record can be proven to be what it purports to be, to have been created or sent by the person that created or sent it and to have been created or sent at the time it is purported to have occurred<br>Source: INTERNATIONAL COUNCIL ON ARCHIVES, 2008, <i>Principles and functional requirements for records in electronic office environments</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| MoReq specifications | Authenticity, principle of –<br>Definition: Along with integrity, reliability and usability, one of the central characteristics of a record according to ISO 15489.<br>An authentic record is one that can be proven to be what it purports to be. Source: DLM Forum Foundation, 2011. Version 1.1: MoReq 2010&reg;: modular requirements for records systems - volume 1: core services & plug-in modules [en ligne]. S.l. [Consulted the 14th January 2015]. Available at: <a href="http://moreq2010.eu/pdf/moreq2010_vol1_v1_1_en.pdf">http://moreq2010.eu/pdf/moreq2010_vol1_v1_1_en.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Table AI.

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**About the author**

Since 2010, Basma Makhoulf Shabou is a Professor at the Geneva School of Business Administration (HEG-Geneva), the University of Applied Sciences of Western Switzerland, where she teaches in the records management and archives area in the Information Studies Department. From 1998 to 2003, she held the position of Senior Administrator of Public Records and Archives at the National Archives of Tunisia. In 2011, she obtained her PhD in information sciences (available at: <https://papyrus.bib.umontreal.ca/xmlui/handle/1866/4955>) from the University of Montreal (Canada), where she also taught courses in archival sciences and records management. She has also a license in social studies and a postgraduate degree in records management. Makhoulf Shabou has been lecturing at a number of archival schools across Tunisia and Canada. Since 2013, she joined Records in the Cloud ([www.recordsinthecloud.org/ritc/home](http://www.recordsinthecloud.org/ritc/home)) and InterPARES Trust projects ([www.interparestrust.org/trust/aboutus](http://www.interparestrust.org/trust/aboutus)). Her principal research areas focus on defining and measuring the quality of documents; accessibility of documents; relevance and representativeness of archives; archival appraisal and the value of documents; data quality measurement; cloud computing and archival practices; information governance policy and its implementation tools. Basma Makhoulf Shabou can be contacted at: [basma.makhoulf-shabou@hesge.ch](mailto:basma.makhoulf-shabou@hesge.ch)

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